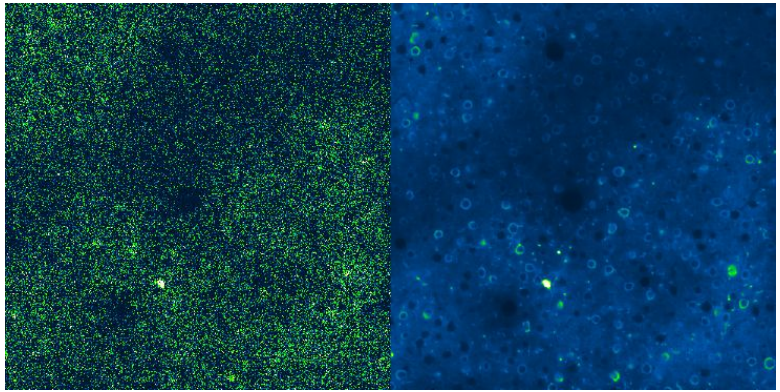
Grand Challenge – AI4Life Calcium Imaging Denoising Challenge 2025

2026/02/02

Group 2
張晉瑋、楊明勳、張維宇、鄭育佳、高翊庭、陳冠璇

Contest Background

- Calcium Imaging (鈣離子成像)
  - 螢光顯微鏡
  - 基因編碼鈣指示劑 (Genetically Encoded Calcium Indicators, GECIs)
  - 觀察細胞中鈣濃度的變化，就能知道細胞的動作。
- AI4Life
  - 由歐盟 Horizon Europe 計畫資助的專案。專門為「光學顯微成像」與「生命科學」量身打造的「 AI 工具資源庫」。



Challenge

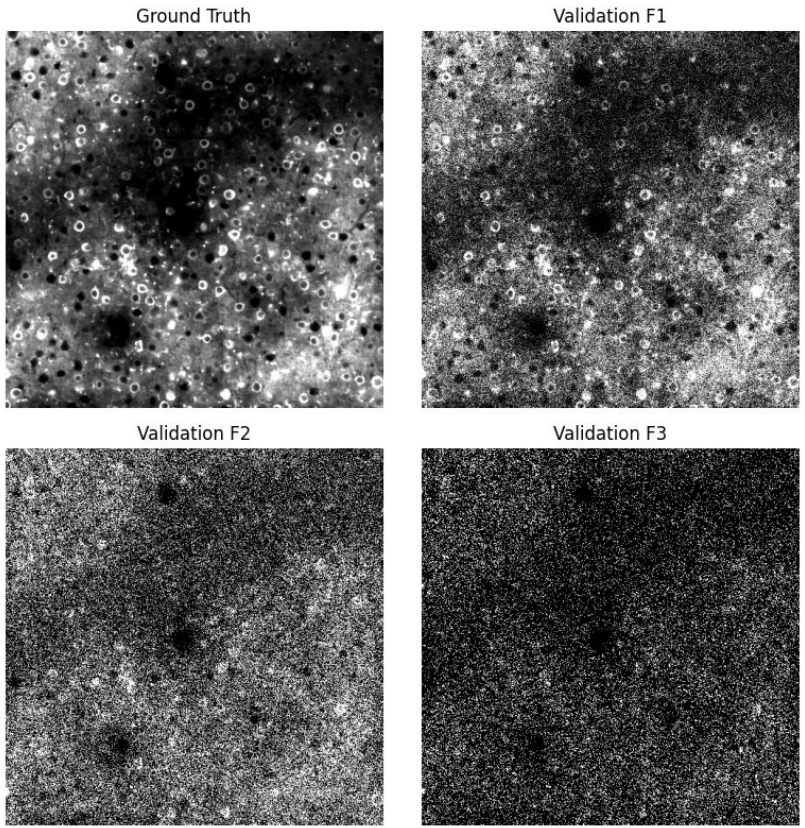
- Why Denoising in Calcium Imaging Is Necessary?
  - 在鈣成像中，研究人員拍到的影像幾乎一定含有大量雜訊，雜訊會讓真正的細胞活動訊號 **變得模糊、被誤判** 。
  - 去雜訊方法必須在保留細胞空間結構（形狀、位置）的同時，也保留時間訊號的動態特性，否則容易將真實的細胞活動一併抹除。
- Dataset Note
  - 鈣成像資料不是一張照片，而是空間和時間的維度：**空間包括細胞的形狀、位置、亮暗分布，而時間則代表同一個細胞隨時間亮度如何變化** 。
  - 在真實實驗中幾乎無法取得完全乾淨的「真實標準答案（ ground truth）」。即使在高訊雜比條件下取得的影像，殘留的雜訊仍可能使傳統的品質評估指標不 夠嚴謹。因此競賽採用 **合成資料集** 作為核心，這些資料中真實訊號是已知的，且雜訊可以被有控制地加入，以模擬真實情境。

Dataset Overview

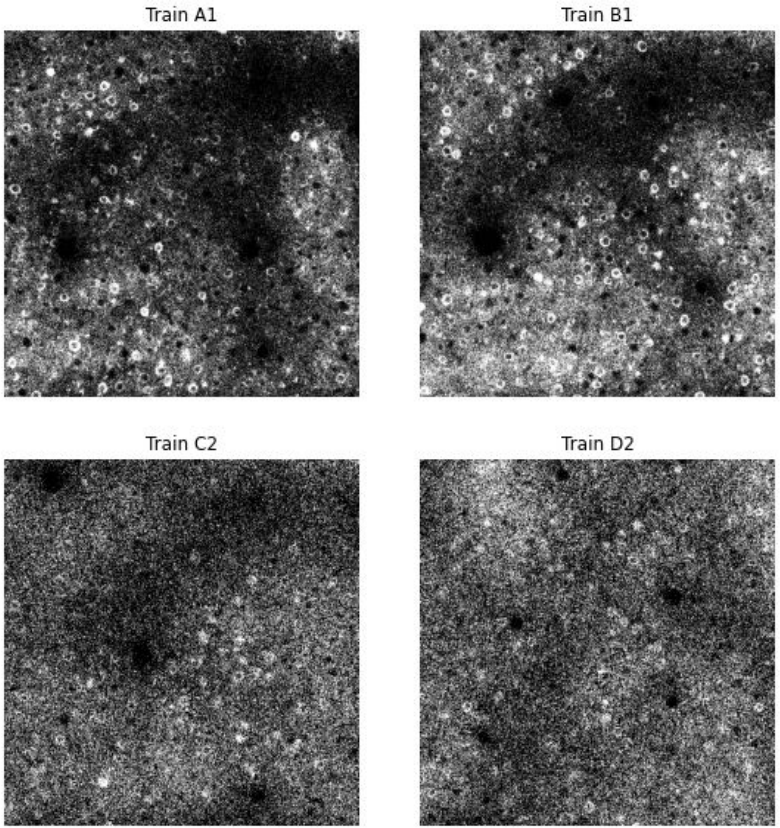
- Dataset format
  - TIFF file (Lossless Compression) in 16-bit gray scale (=65,536 brightness steps)
  - (video, time_frame, width, height)
- Train Dataset (4, 1500, 490, 490)
  - 4 different noised calcium videos
- Validation Dataset (4, 1500, 490, 490)
  - 1 clear ground truth, 3 different noised levels
- Official Test Dataset
  - Task 1: Content Generalization Submission (8, 500, 128, 128)
    - 8 different noised calcium videos
  - Task 2: Noise Level Generalization Submission (4, 500, 128, 128)
    - 4 different noised calcium videos

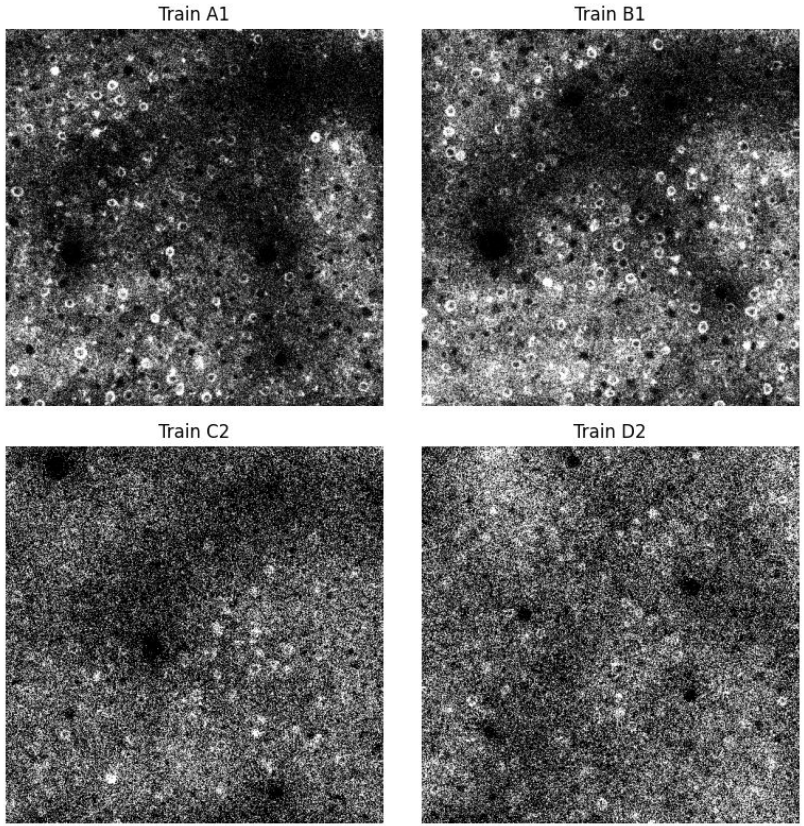
Brightness of Images

- White: Higher brightness → Neuron is active
- Black: Lower brightness → Background or inactive
- Noise
  - Photon Counting (光子計數)
  - Dark Noise / Thermal Noise
- 去噪的目標就是讓「背景更黑(去除雜訊)」，同時讓「訊號的亮度變化(閃爍)」保持真實，不能因為去噪而讓原本該亮的點變暗。

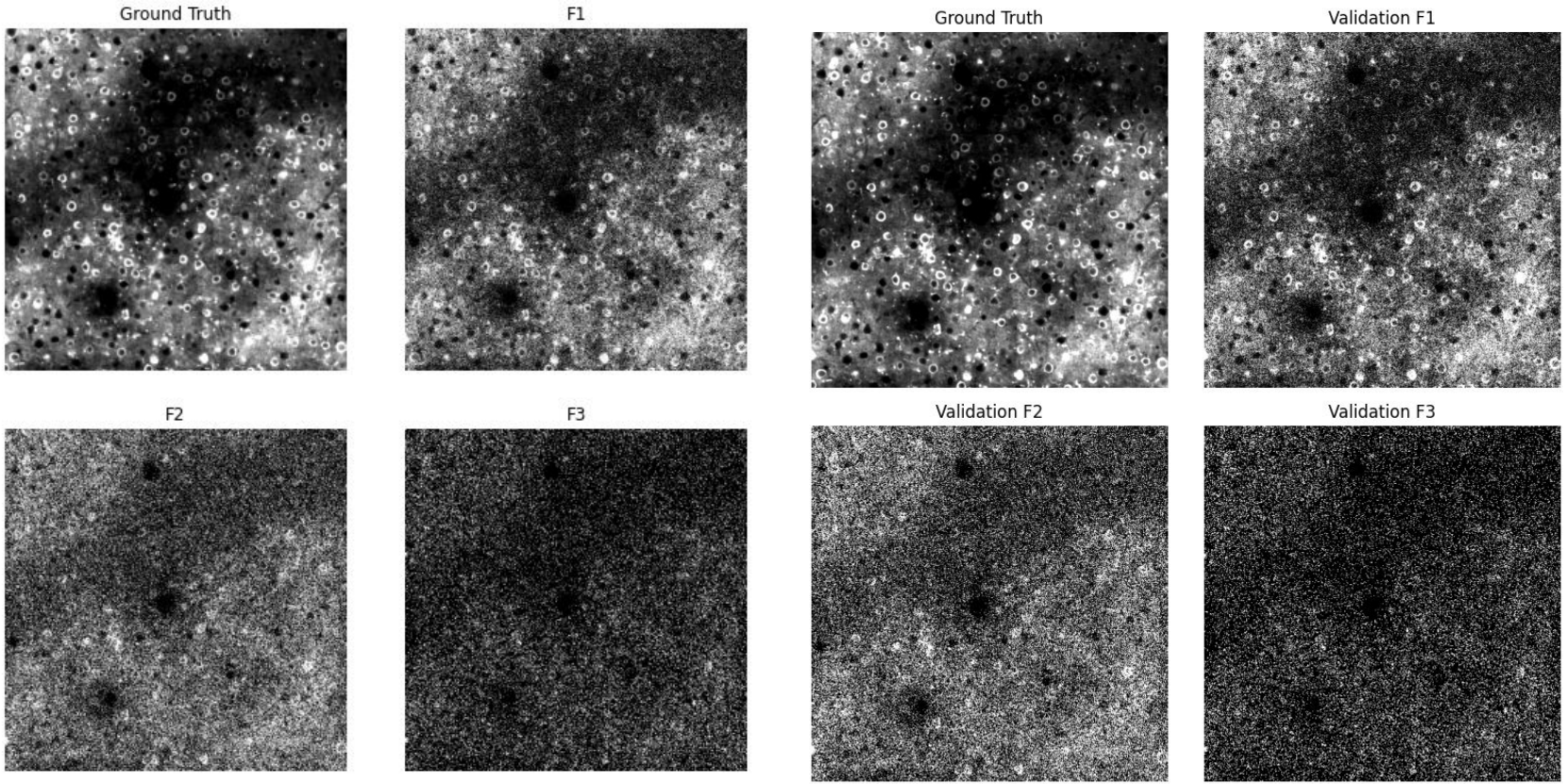


Data Demostration – Train Dataset

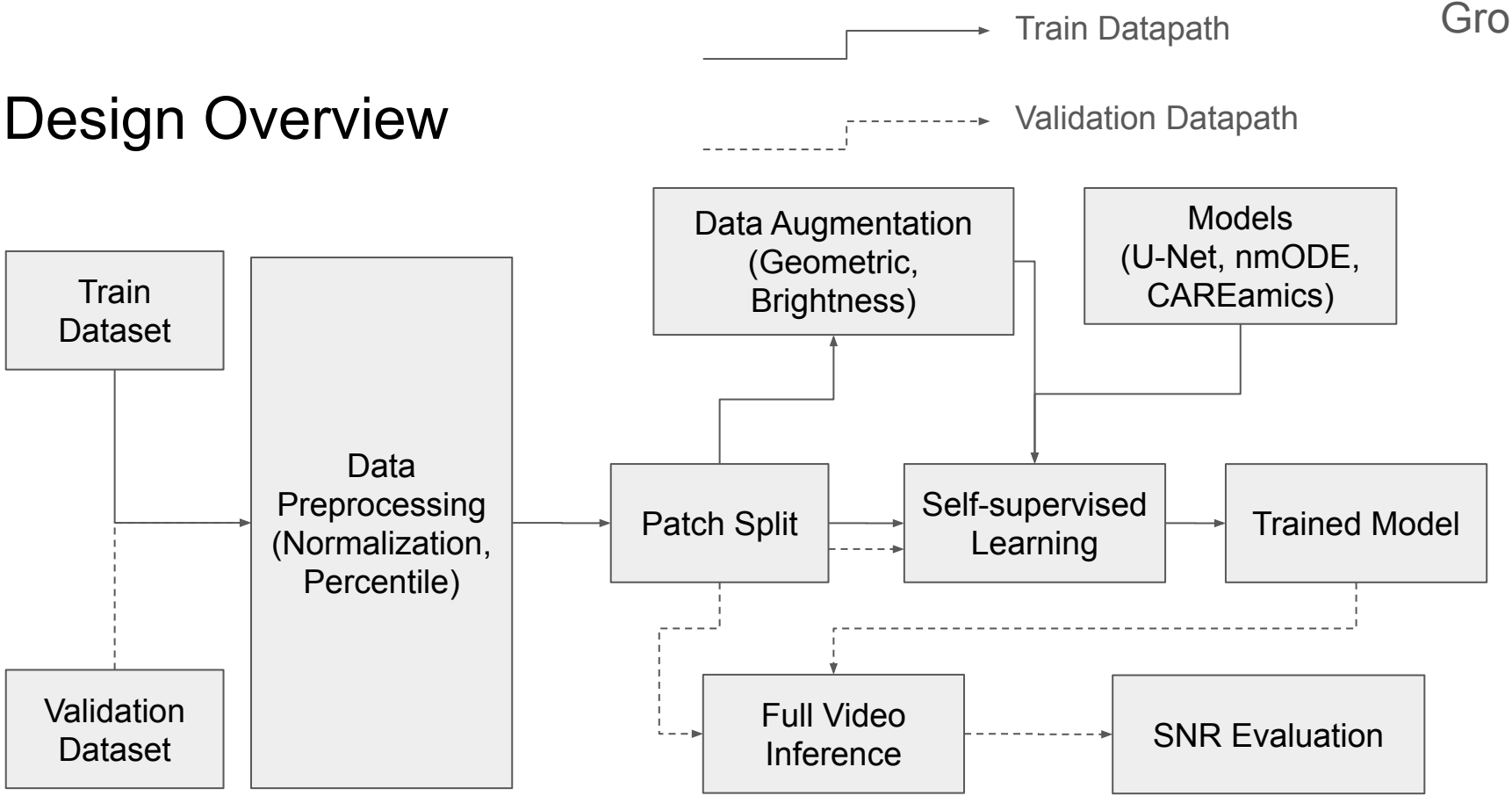




Data Demostration – Validation Dataset

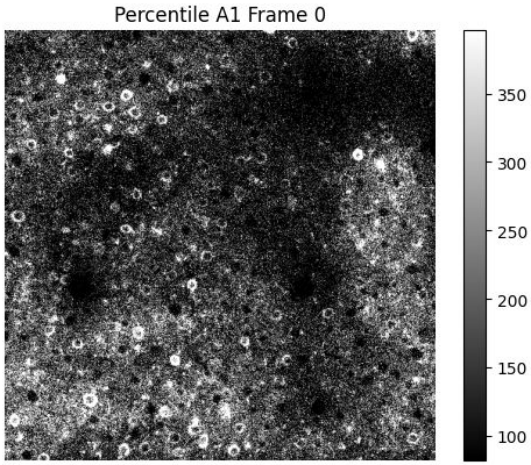


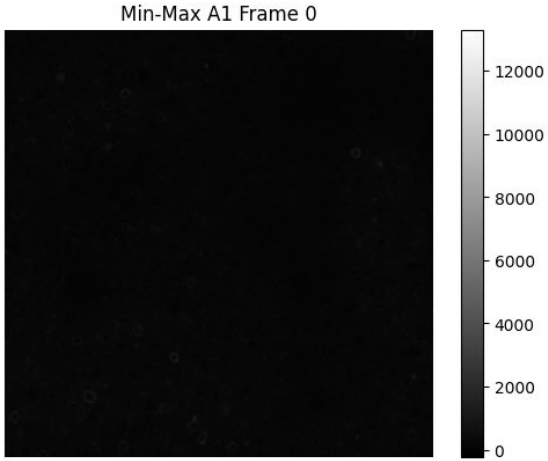
Design Overview



Data Preprocessing

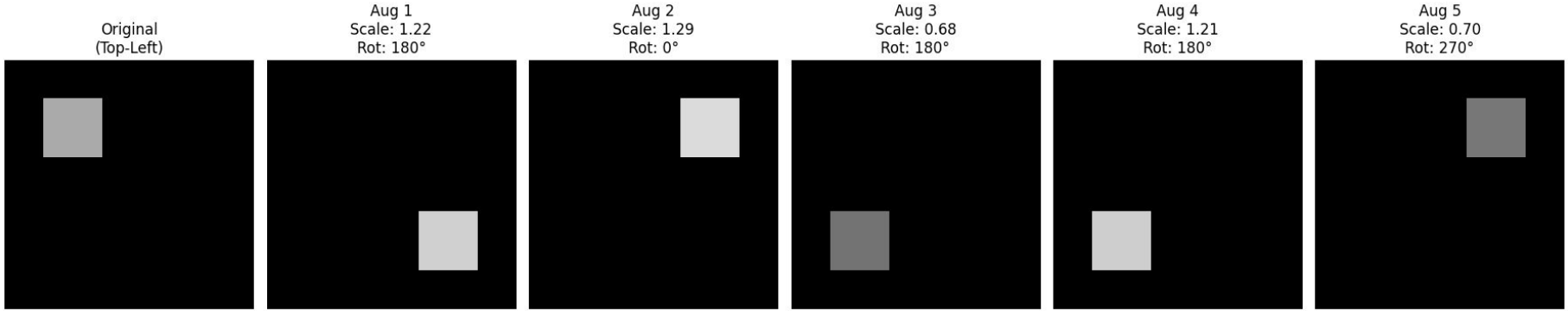
- Normalization
  - Global normalization for train dataset
  - Global normalization for validation dataset
- Percentile at 1th, and 99.9th
  - train dataset → -89, 1275
  - validation dataset → -100, 2090
- Why not use min, max? Remove outliers!
  - For A1 image
    - max: 13306
    - min: -247





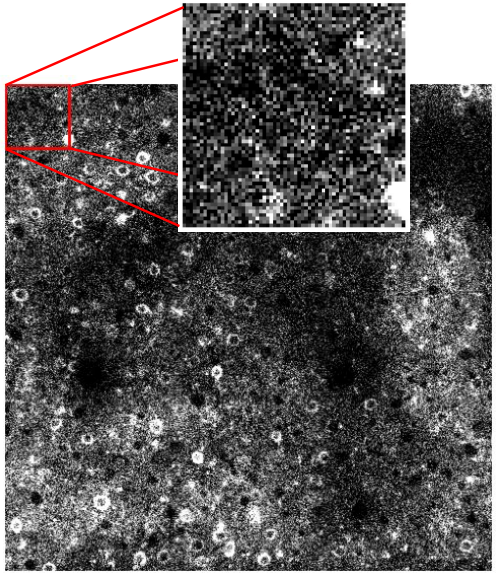
Data Augmentation – Geometric and Brightness

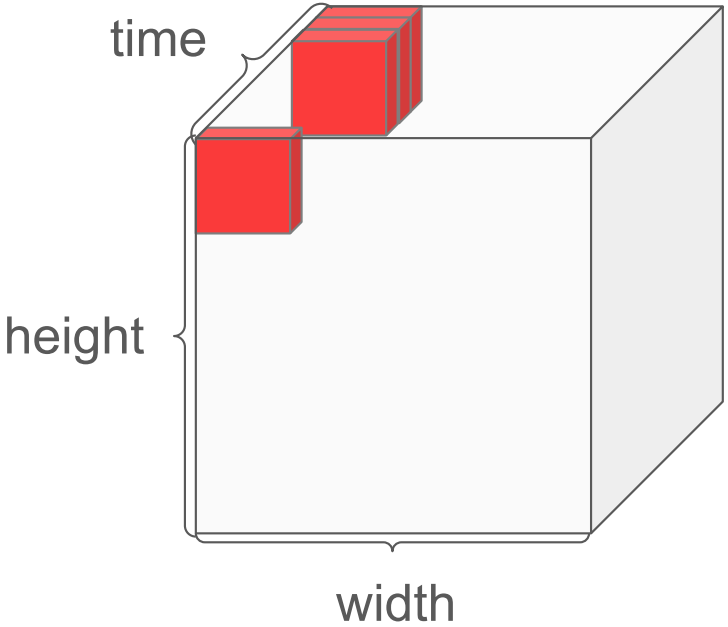
- Geometric
  - Mirror
  - Rotation 90, 180, 270
- Brightness
  - original_value * uniform(0.8, 1.4)



Patch Training

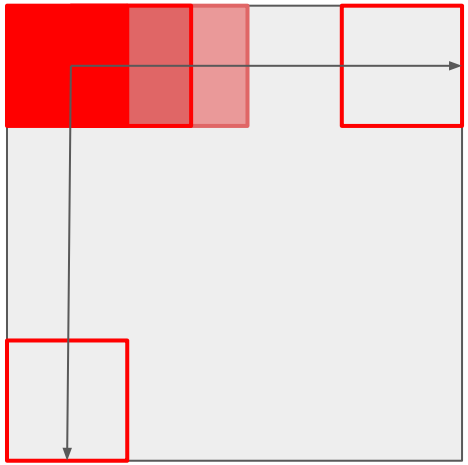
- Inference whole frame is hard and costly
- Split a video to 32 * 64 * 64 patch size for training and inference





Full Video Inference

- Sliding Window
- Accumulation and Average
  - accumulation_map
  - counting_map



assume 2x2 patch size

accumulation map

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16

counting map

1	2	2	1
2	3	3	2
2	3	3	2
1	2	2	1

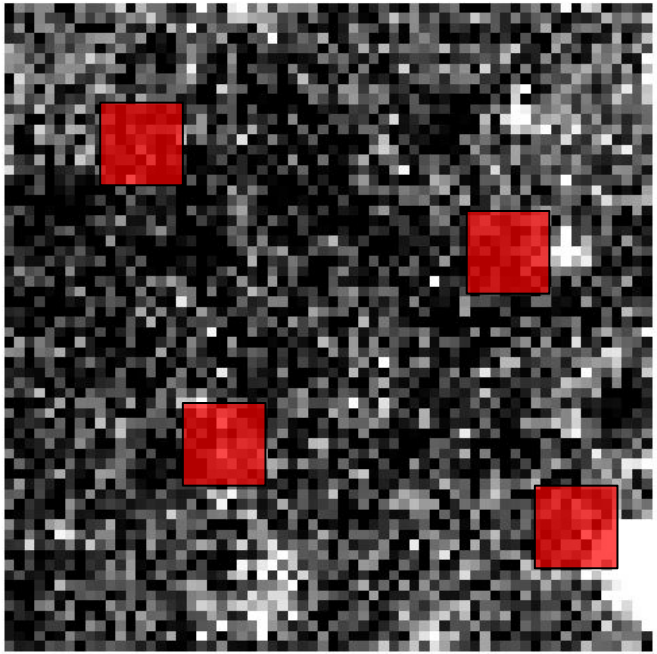
result map

1	1	3/2	4
5/2	2	7/3	4
9/2	10/3	11/3	6
13	7	15/2	16

Self-Supervised Learning: Masked Image Training

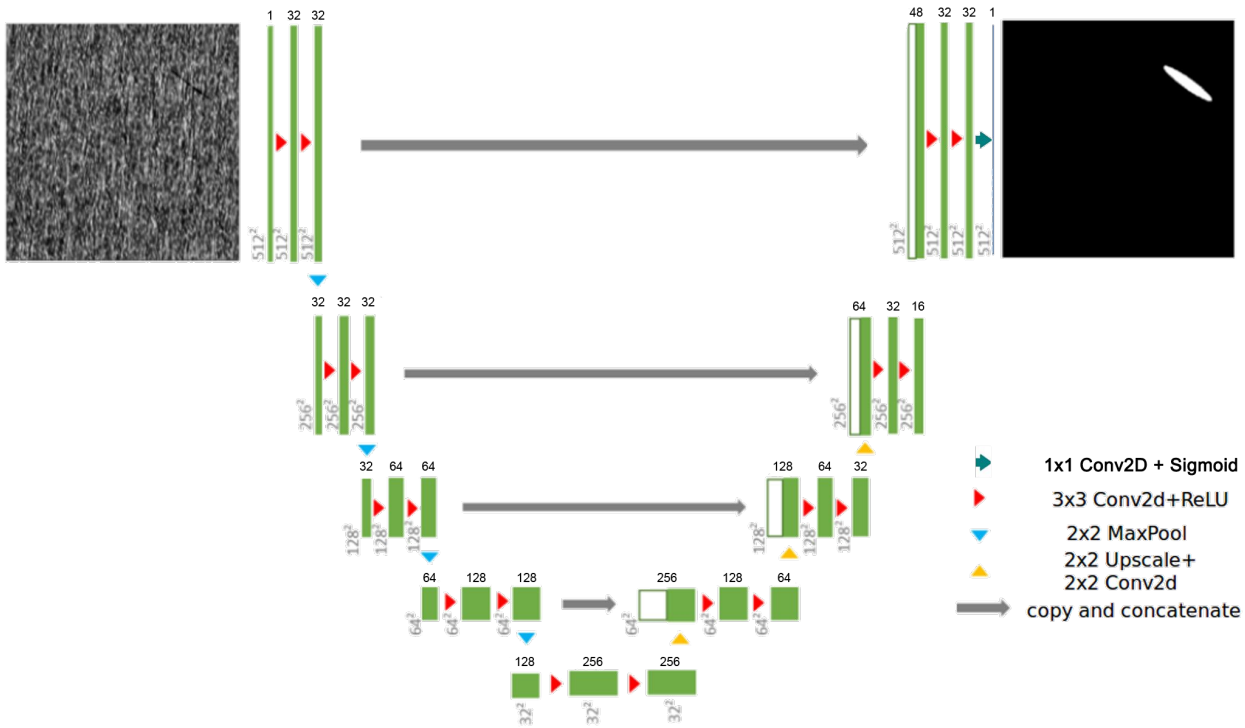
- Gaussian Noised Masking
  - calculate patch range std and mean, and generate noisy to fill mask part

= mask part



Baseline Denoised Solution: U-Net

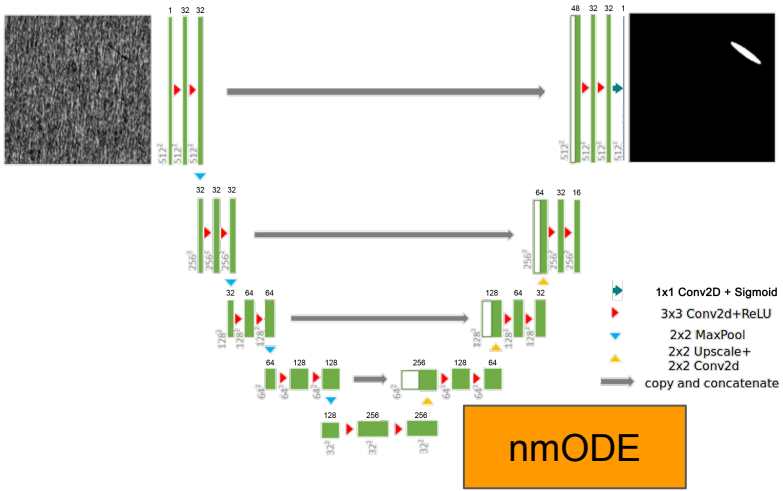
- Encoder
  - 負責看特徵
- Bottleneck
  - 理解特徵
- Decoder
  - 還原與重建圖片
- Skip connection
  - 幫助復原細節



ref: https://catalog.ngc.nvidia.com/orgs/nvidia/teams/dle/resources/unet_2d-industrial_tf1?version=20.06.0

Temporal-Aware Solution: 3D U-Net & nmODE

- We should aware the time relation. → 3D U-Net
  - 3D → (time, height, width)
- neural memory Ordinary Differential Equation (nmODE)
  - nmODE block location → after bottleneck
  - nmODE effect
    - 時間連續性記憶體



[1] Yi, Z. nmODE: neural memory ordinary differential equation. Artif Intell Rev 56, 14403–14438 (2023). https://doi.org/10.1007/s10462-023-10496-2
[2] Chen, Ricky TQ, et al. "Neural ordinary differential equations." Advances in neural information processing systems 31 (2018).

Experiment Setup

- Comparison
  - 2D U-Net
  - 3D U-Net
  - nmODE U-Net
  - CAREamics N2V U-Net
- Computation Resources
  - Colab
  - Kaggle
  - RunPod
- Code
  - https://github.com/jinwei-chang/grand-challenge-ai4life-cidc-2025

Evaluation Metrics

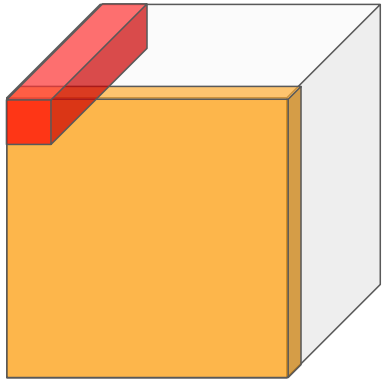
- Signal-to-Noise Ratio (SNR)
- Spatial SNR (sSNR)
- Temporal SNR (tSNR)
- Saptio-Temporal SNR (stSNR)
- Final Score (avg stSNR)

$$sSNR(x,y) = \frac{1}{T}\sum_{t=1}^{T} 10\log_{10}\frac{||y_t||^2}{||x_t - y_t||^2}$$

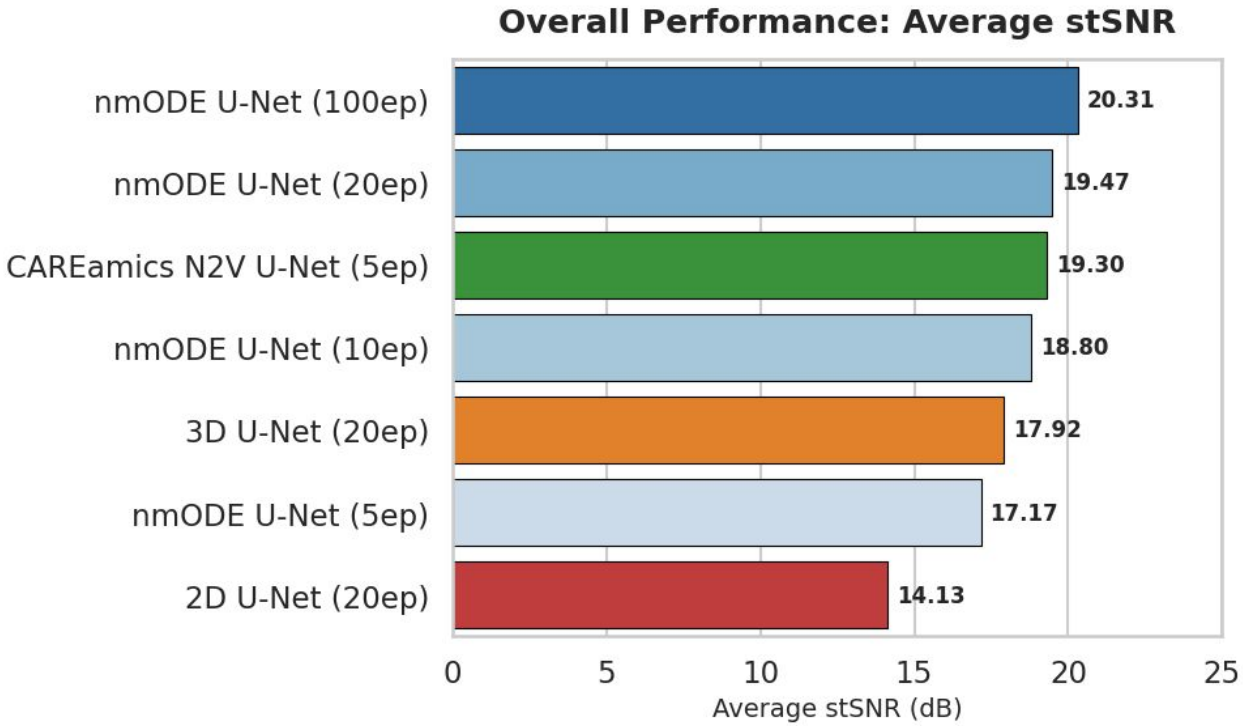
$$tSNR(x,y) = \frac{1}{HW}\sum_{i=1}^{W}\sum_{j=1}^{H} 10\log_{10}\frac{||y_{i,j}||^2}{||x_{i,j} - y_{i,j}||^2}$$

$$stSNR(x,y) = \alpha \cdot sSNR(x,y) + (1-\alpha)\cdot tSNR(x,y), \alpha \in [0,1]$$

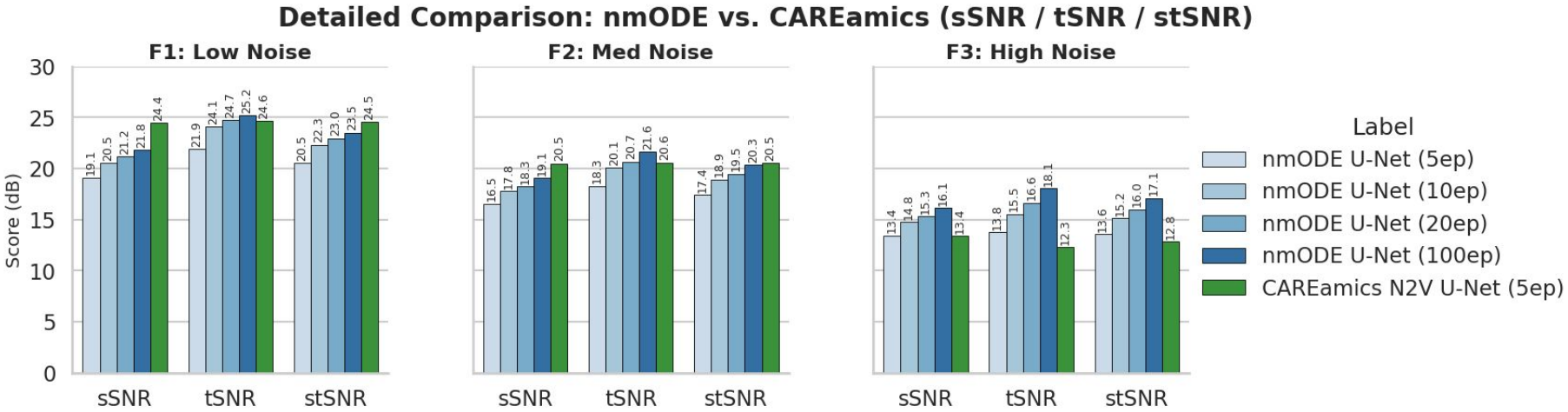
$$score = \frac{1}{F}\sum_{f=1}^{F} stSNR(x_f, y_f)$$



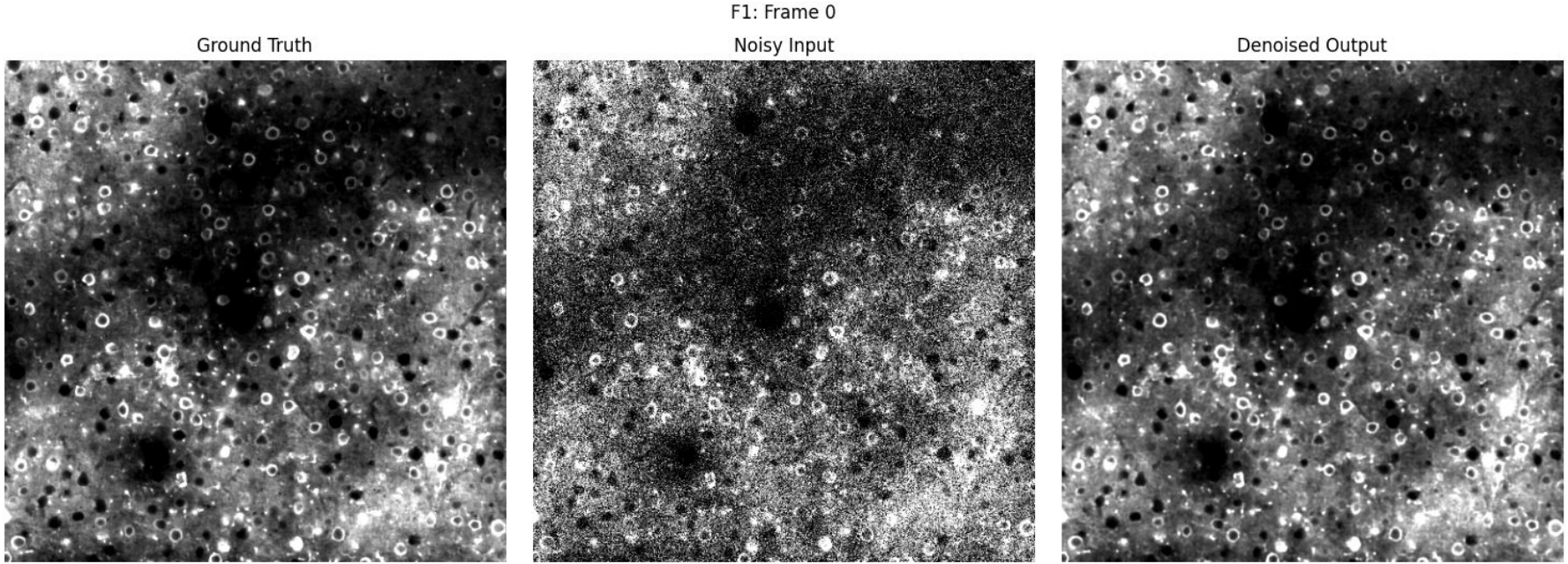
Final Score (Average stSNR) Evaluation



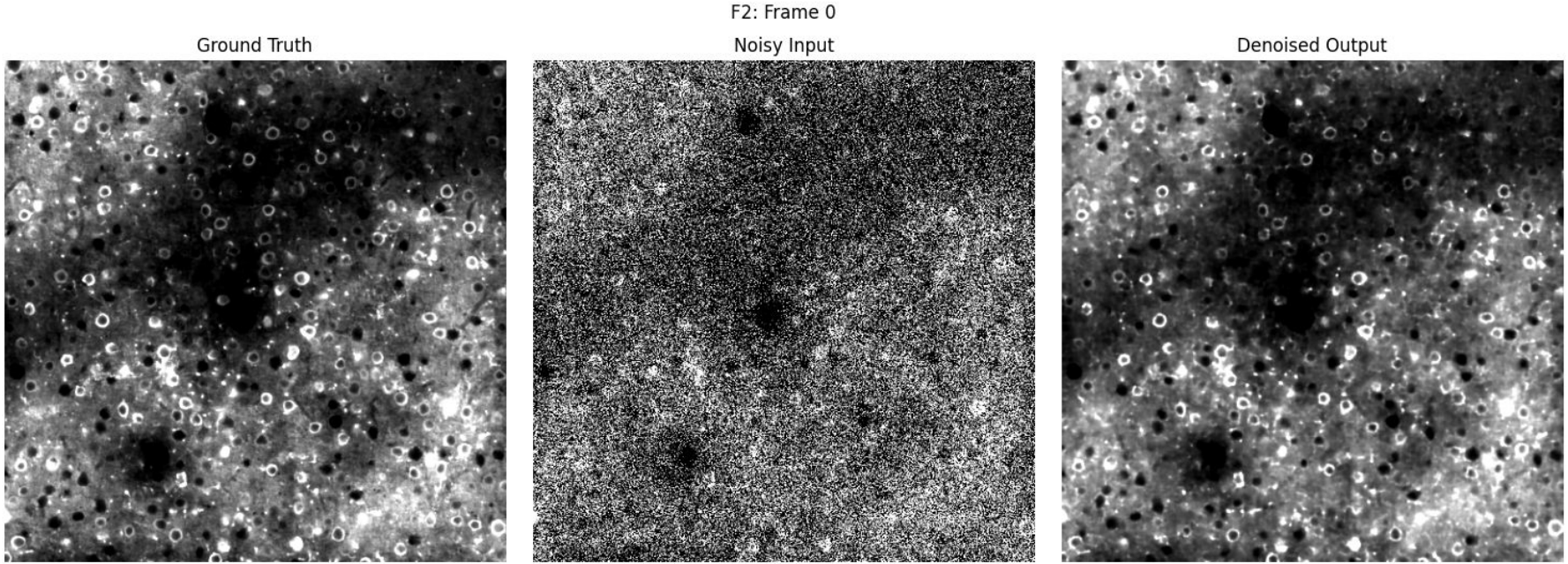
sSNR, tSNR, and stSNR



Denoised Demonstration – F1



Denoised Demonstration – F2



Denoised Demonstration – F3

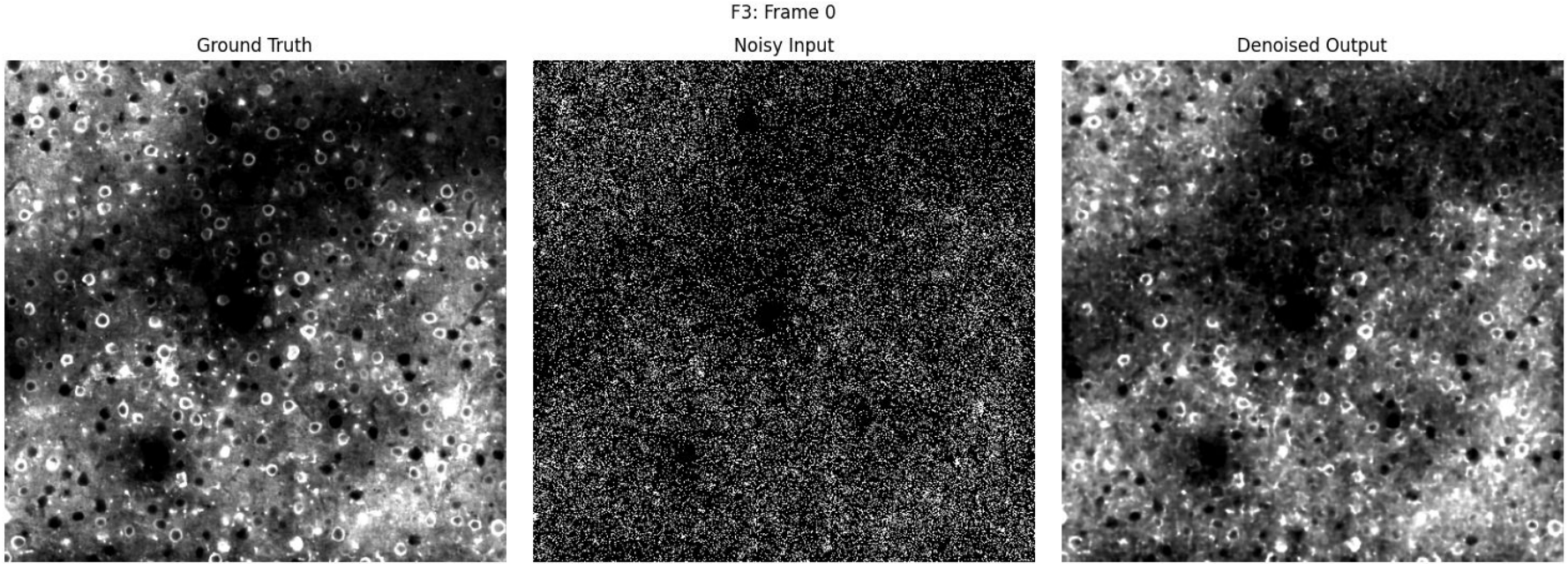


Image Histogram Demonstration – F1

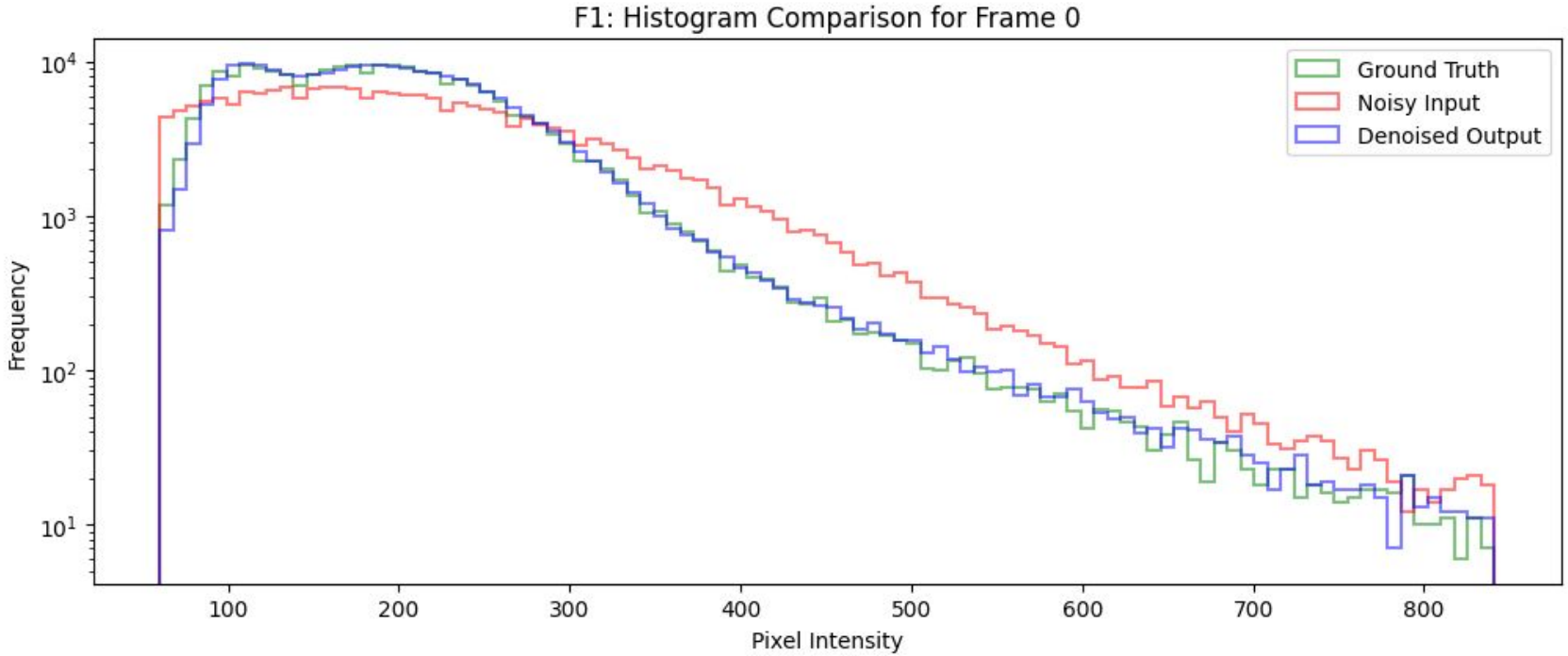


Image Histogram Demonstration – F2

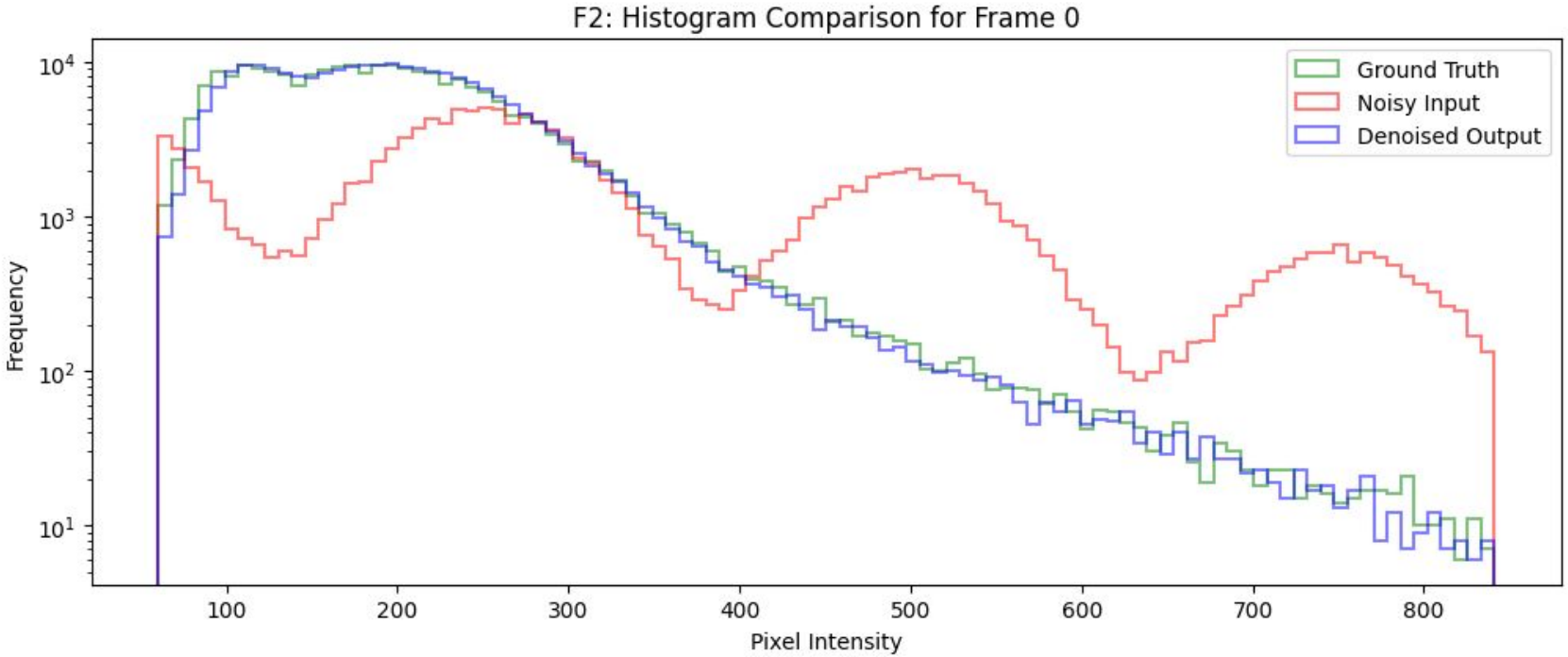
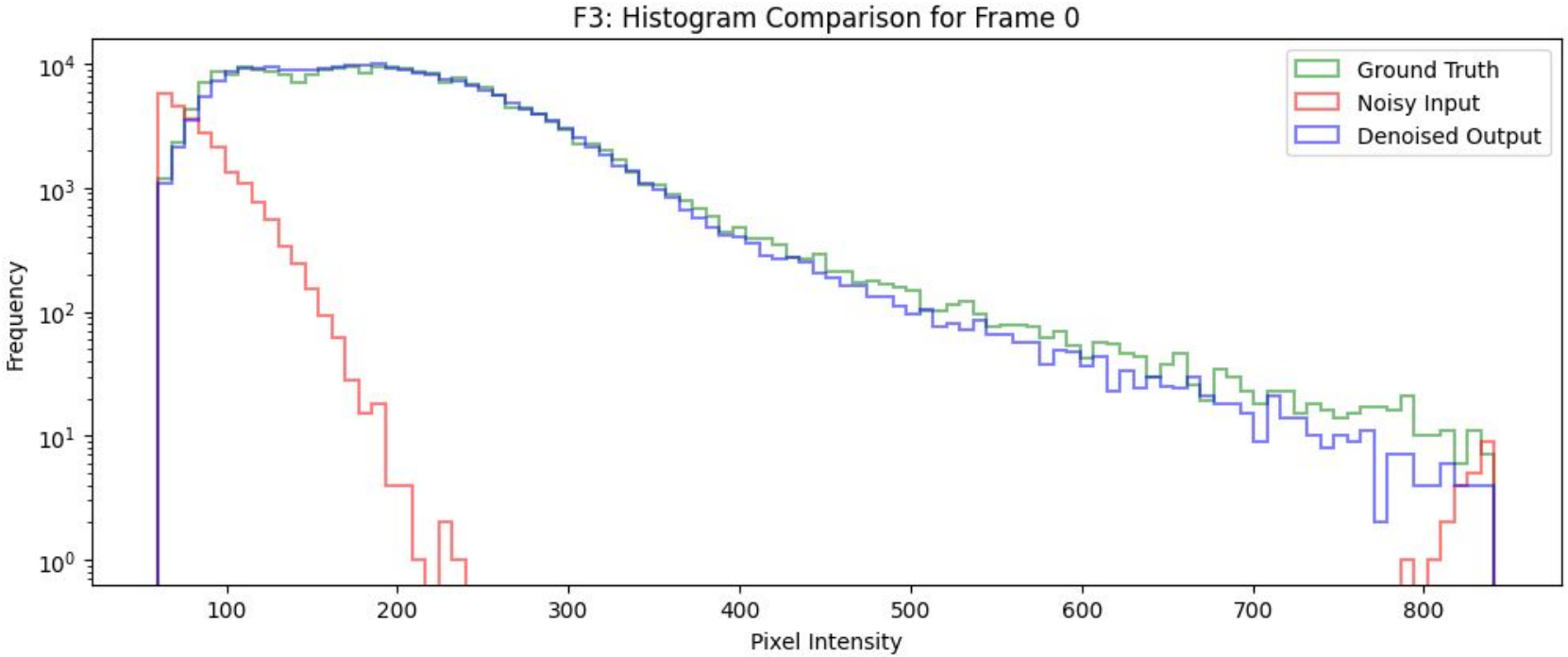


Image Histogram Demonstration – F3



Future Work: Official Test Submission

Final Submission Phase: Noise Level Generalization Leaderboard

Search:

Additional metrics | Show all metrics

#	User	Algorithm	Created	Score	Comment	Publication
1st	edoardogiacomello	N2V 3D	22 Aug. 2025	16.75	N2V_3DNB_V1 No-ConvBias Patch Size 64x64x64	
2nd	chhayanshporwal	AI4Life CIDC25 Submission	21 Jan. 2026	15.59		
3rd	mcroft	CIDC n2n windowed	31 Aug. 2025	15.10	N2N 2D windowed	
4th	edoardogiacomello	N2V 3D	22 Aug. 2025	14.93	N2V_3DNBNA_V1 No-ConvBiasNoBNAffine Patch Size 64x64x64	
5th	edoardogiacomello	N2V 3D	22 Aug. 2025	9.99	N2V2_3D_V1 Patch Size 64x64x64 Batch Size50	
6th	edoardogiacomello	Baseline	3 July 2025	8.75		
7th	Guillermo_Rey	NafNet2Void 2D	23 Aug. 2025	6.35		
8th	Guillermo_Rey	NafNet2Void 2D	24 Aug. 2025	6.24		
9th	edoardogiacomello	N2V 3D	21 Aug. 2025	5.57	N2V_3D_V1 Patch Size 64x64x64 BS50	

Conclusion

- We applied nmODE U-Net to address the temporal instability in the Calcium Imaging Denoising Contest.
- Our method nmODE U-Net is superior to CAREamics in terms of temporal stability (tSNR).
- At the highest noise level (F3), all metrics (sSNR, tSNR, and stSNR) from nmODE U-Net are better than CAREamics.
- In the future, we will submit our method to the official evaluation system to get a fair benchmark.